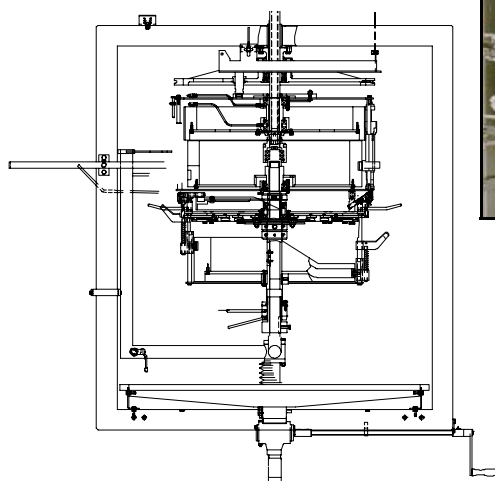
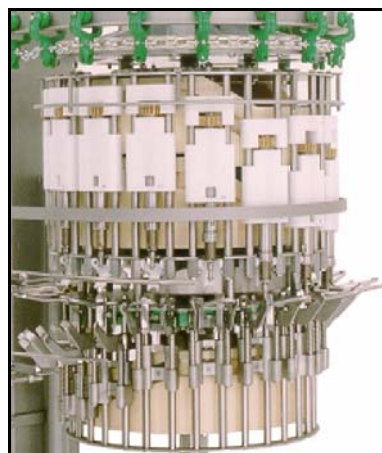
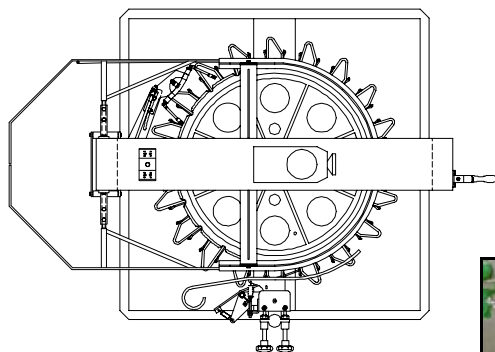


User- and Maintenance manual



0100-5

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Foreword

Meyn Food Processing Technology B.V. is one of the worlds leading designers and manufactures of systems and equipment for the poultry processing industry. Meyn's flexible service and support organisation and its constant emphasis on product innovation through extensive research and development, enable Meyn to supply today's customers with the solutions of tomorrow.

Regardless of the size or complexity of the process, Meyn will supply its customers with situation specific solutions that are critical in reaching their objectives. Therefore, Meyn combines a solid knowledge of the customers business with expert process design skills. This is crucial, as only well focused solutions generate the desired returns.

Our ambition is to supply solutions that support fully controlled in line processing; handling and tracking every product or batch throughout the entire facility while controlling the quality of the output, combined with the ability to match the incoming birds with the sales orders in the most optimal manner.

Meyn is worldwide represented by a network of subsidiaries and agents. In order to supply complete integrated solutions to our customers, Meyn has a number of strategic partners in the following processing sectors:

- Bonescan.
- Mechanical deboning.
- Logistic systems.
- Refrigeration.
- Water treatment.
- Vacuum transport.
- Rendering.
- Convenience food.

This manual

This manual is intended for the operator(s) of this machine. It's a means for information and instruction, so you can use the machine correctly and safely.

The manual consists of 7 chapters and a few appendices:

- Chapter:1. Introduction, includes general information about the machine and the supplier.
- Chapter:2. Safety, includes a description of the most important safety aspects associated with the use of the machine.
- Chapter:3. Installation and set-up, includes a description of the actions that must be performed before the machine can be used.
- Chapter:4. Operation, the actions that must be performed during normal operation.
- Chapter:5. Cleaning, includes a description of the cleaning activities that must be carried out to keep the machine in good condition and to facilitate safe and hygienic operation.
- Chapter:6. Maintenance, includes a description of the maintenance activities that are necessary to keep the machine in good condition and to facilitate safe operation.
- Chapter:7. Disassembly/assembly and adjustments describes the disassembly and assembly of parts needed by reinstallation of the machine to another place or environment and to ensure that the machine performs safely and efficiently.
- Appendix:. Appendices, includes aspects/details for the machine.

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1 Introduction

This chapter contains technical data about the machine and gives a description of the purpose and working of the machine.

1.1 Technical specifications

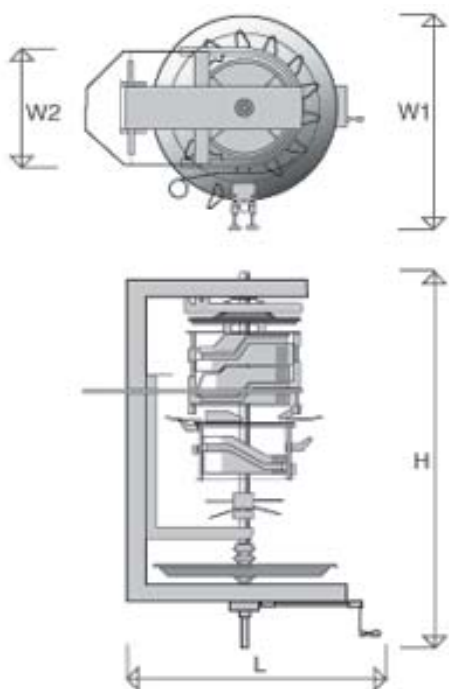


Figure 1.1 Meyn Vent cutter.

Specifications: 7x12"			
Capacity max.	** 3,000 BPH	Height (H)	2,150 mm
Live weight	0.5-3.8 kg	Weight machine	405 kg
Number of units	7	Water connection	3/4"
Shackle pitch	12"	Water consumption	** 0.6 m ³ /h
Length (L)	1,745 mm	Drain connection	DIN100
Width (W1)	1,295 mm	Noise level	70 dB(A)
Width (W2)	680 mm		

Table 1.1 Technical specifications.

<i>Specifications: 11x8"</i>			
Capacity max.	** 4,500 BPH	Height (H)	2,150 mm
Live weight	2.5-5 kg	Weight machine	600 kg
Number of units	11	Water connection	3/4"
Shackle pitch	8"	Water consumption	0.6 m ³ /h
Length (L)	1,745 mm	Drain connection	DIN100
Width (W1)	1,295 mm	Noise level	70 dB(A)
Width (W2)	710 mm		

Table 1.2 Technical specifications.

<i>Specifications: 14x6"</i>			
Capacity max.	6,000 BPH	Height (H)	2,150 mm
Live weight	0.5-3.8 kg	Weight machine	600 kg
Number of units	14	Water connection	3/4"
Shackle pitch	6"	Water consumption	0.6 m ³ /h
Length (L)	1,745 mm	Drain connection	DIN100
Width (W1)	1,295 mm	Noise level	70 dB(A)
Width (W2)	680 mm		

Table 1.3 Technical specifications.

<i>Specifications: 15x8"</i>			
Capacity max.	** 7,500 BPH	Height (H)	2,150 mm
Live weight	2.5-5 kg	Weight machine	720 kg
Number of units	15	Water connection	3/4"
Shackle pitch	8"	Water consumption	0.6 m ³ /h
Length (L)	2,335 mm	Drain connection	DIN100
Width (W1)	1,640 mm	Noise level	70 dB(A)
Width (W2)	970 mm		

Table 1.4 Technical specifications.

<i>Specifications: 16x6"</i>			
Capacity max.	** 8,000 BPH	Height (H)	2,150 mm
Live weight	0.5-3.8 kg	Weight machine	650 kg
Number of units	16	Water connection	3/4"
Shackle pitch	6"	Water consumption	0.6 m ³ /h
Length (L)	1,760 mm	Drain connection	DIN100
Width (W1)	1,345 mm	Noise level	70 dB(A)
Width (W2)	780 mm		

Table 1.5 Technical specifications.

<i>Specifications: 20x6"</i>			
Capacity max.	** 10,000 BPH	Height (H)	2,150 mm
Live weight	0.5-3.8 kg	Weight machine	850 kg
Number of units	20	Water connection	3/4"
Shackle pitch	6"	Water consumption	0.6 m ³ /h
Length (L)	2,335 mm	Drain connection	DIN100
Width (W1)	1,640 mm	Noise level	70 dB(A)
Width (W2)	970 mm		

Table 1.6 Technical specifications.

1.2 General arrangement drawing

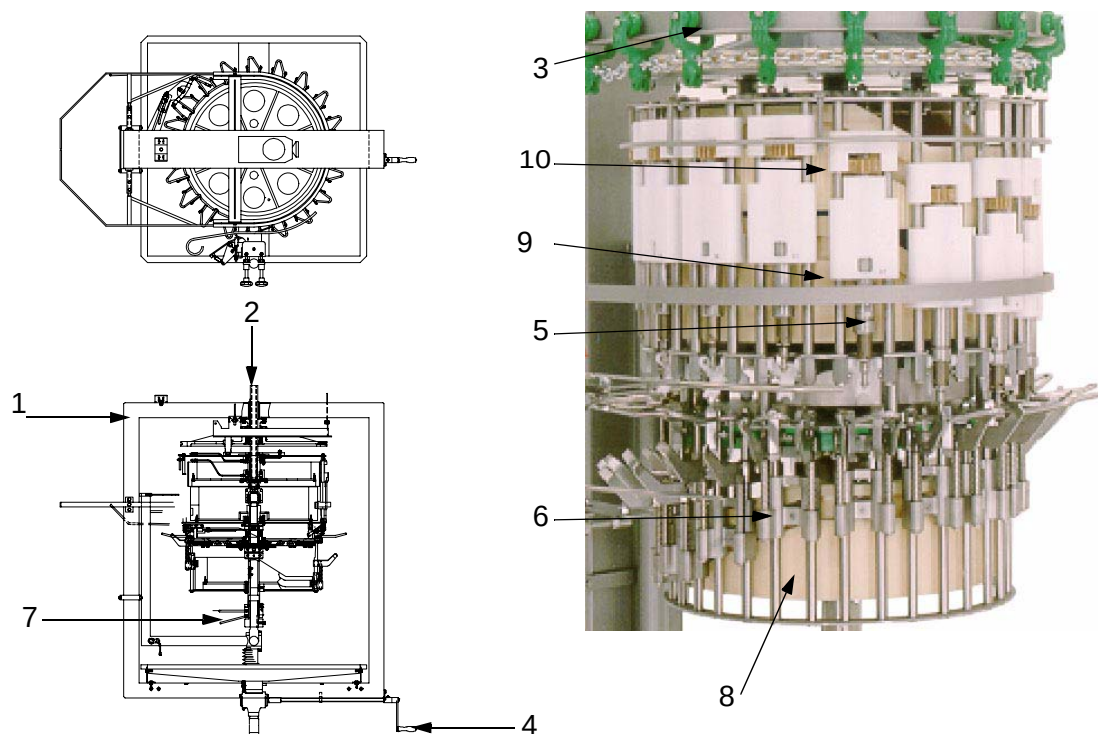


Figure 1.2 Meyn Vent cutter outside and inside detail.

Meyn Vent cutter

The machine consists of the following main components:

- Evisceration frame.
- Main shaft.
- Drive wheel (180° bend).
- Hand wheel.
- Drill unit.
- Lift-up unit.
- Capstan wheel.
- Bottom cam.
- Tail clip cam.
- 1. Top cam.

1.3 What is a Meyn Vent cutter

The vent cutter is developed for use in an eviscerating line. After the slaughter department, the vent and the bursa of the bird are removed. Removing the vent and the bursa is a processing step to prepare the bird for Meyn open cutter.

The purpose of the Meyn vent cutter is removal of the vent and the bursa and hang them over the back of the bird.

The machine is driven by the overhead conveyor (A). The birds, that hang in the shackles of the overhead conveyor by their feet, are lead through the machine. Inlet guides lead the birds into the correct position of the machine.

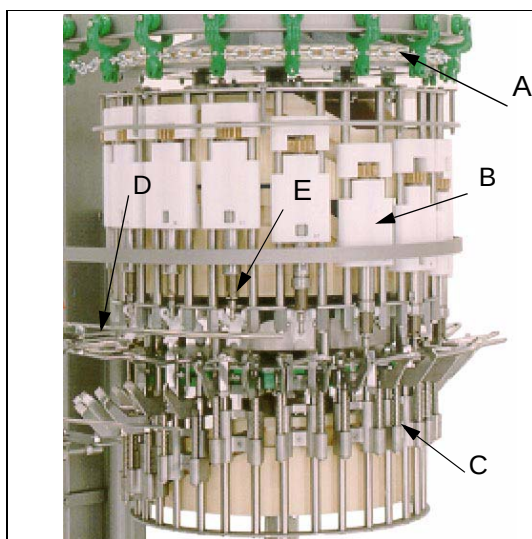


Figure 1.3 Detail (inside the Meyn Vent cutter)

Five cams (top cam, bottom cam, tail clip cam, back plate cam and knife unit cam) in which a curve path is milled or grooves are milled drive the processing units.

Birds are guided into a processing units. A processing unit consist of an upper part (B) and a lower part (C).

The lower part lifts the bird up and pushes it against a centre bracket (D).

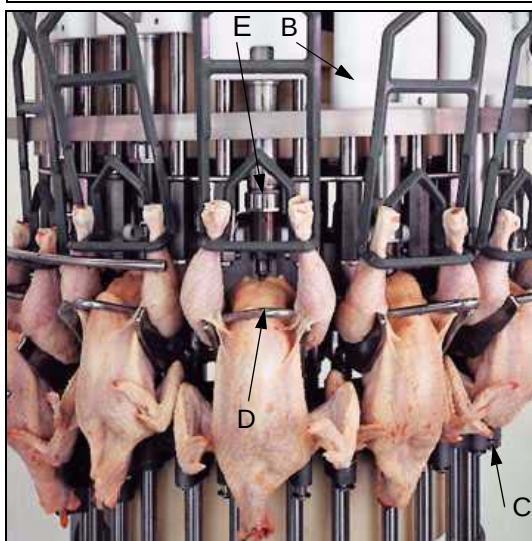


Figure 1.4 Detail (inside the Meyn Vent cutter)

The upper part contains a drill mechanism (E), that drives a drill rod. The drill rod makes a downwards rotary motion into the bird. A vent knife cuts the vent and the bursa of the bird.

After this the drill rod is retracted and hang the vent and bursa over the back of the bird.

The bird is now taken to the next processing step.

1.4 The end product

The output from the machine must satisfy the following requirements: (dependents o.a. on the proceeding bird type.)

- Bursa and vent hung over the back of the bird.
- Bursa completely removed.
- No damage to the intestines.
- Back or kidneys not damaged.
- Hips not cut.
- High yield.

1.5 The Meyn Vent cutter technical

The machine is directly driven by the overhead conveyor. The machine is made of stainless steel and other approved materials for food processing.

The height of the machine is adjustable by using a handwheel. The height of the curve is adjustable by using a capstan wheel.

A Torque switch is a machine safety application to prevent the machine from broken machine parts in case of disasters (if applied).

1.6 Performance of the machine

The Meyn vent cutter is suitable for various types of poultry. The machine gives a continuous performance if the machine is maintained and adjust according to the given directives.

2 Safety

This chapter contains a description of safety aspects relating to the machine. It is particularly important that you are fully aware of the contents of this section. The Meyn Food Processing Technology B.V. has taken all possible steps to provide you with all the necessary information concerning any dangers that may exist when the machine is used. The customer is responsible for understanding and applying these safety rules.



If any of these aspects are not clear, please ask the manufacturer for an explanation.

2.1 The most important safety risks

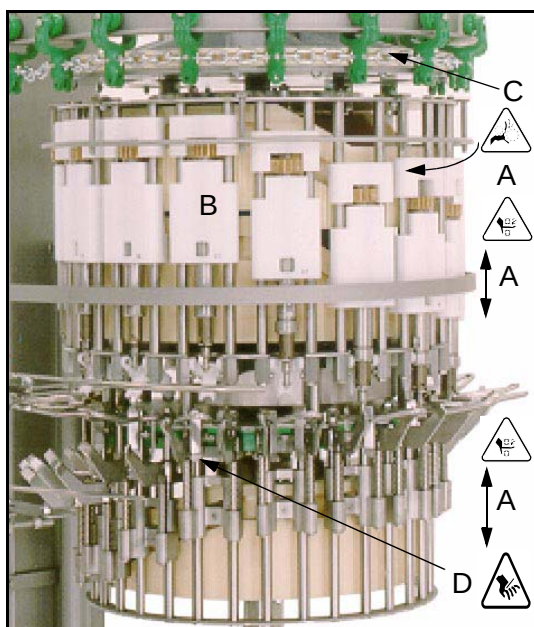


Figure 2.1 Safety risks.

- Moving parts (A). A hand or a garment can be stuck in the moving parts inside the machine.
- Between the rods and units (B), drive chain and drive wheel you can get jammed (C).
- The vent knives are (D) very sharp, that might cause injury in case of incorrect use. Do not try to remove any birds or bird parts from the working vent knives.

2.2 Safety devices (if applied)

In order to make the use of this machine as safe as possible the following safety devices have been fitted:

- The design is an so-called “open” construction without protecting covers.

There is an emergency stop pull cord around the machine. This may only be operated in an emergency. If an operator is accidentally caught up in the machine or falls into the machine or intentionally operates it manually, the emergency stop will automatically activate and stop the entire machine. You can also activate the emergency stop by pushing the red push-button. The emergency stop pull cord and possibly the red push-button will then have to be reset.

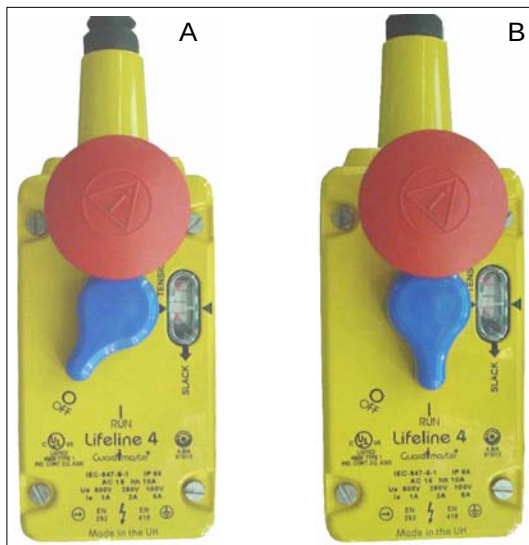


Figure 2.2 The emergency stop pull cord.

Functioning

- The blue switch is switched automatically to “0; OFF” (A) when the pull cord has been activated. See figure.
- Check the tension of the pull cord. See “Tension of the pull cord” and figure below.
- Check that the emergency situation has been resolved.
- Switch the blue switch to “RUN” (B) to reset the emergency stop pull cord. See figure.

- Reset the emergency stop switch circuit by means of pushing the lighted blue reset emergency stop button on the main panel. After this the light in the button goes off.
- Accept the fault on the touch screen.
- After this you can restart the machine.

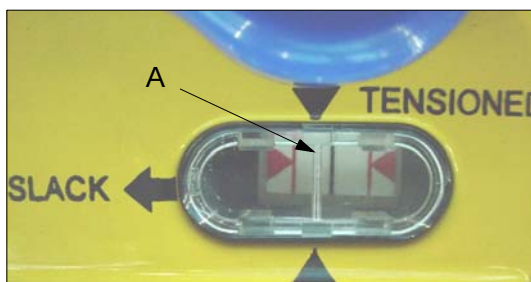


Figure 2.3 The pull cord tensioner window.

Tension of the pull cord

- The stripe (A) must be in the middle of the window.
- If necessary change the tension of the cord by turning the cord tensioner.

2.3 Safety measures to be taken

Before and during use of the machine it is important to take a number of safety measures and instructions into account, which are described in the sections below.

2.3.1 Preparing for safe operation

Before operating the machine make sure that you:

- First read the relevant instructions in this manual.
- Wear footwear which prevents slipping.
- Don't wear loose-hanging clothing.
- You must take note of all the safety symbols attached to the machine.
- Check that the power supply complies with the locally applicable regulations.
- Ensure that the machine is correctly placed in a stable position.
- Ensure that the operation, minor maintenance activities and activities concerned with installing or moving the machine are only performed by qualified personnel, that is to say by personnel who have the necessary skills and are familiar with the contents of this manual.
- Modify the machine in any way, will endanger the safe and proper functioning of the machine.

2.3.2 Safety measures during operation

For safe operation of the machine, it is essential that a number of safety instructions are taken:

- Ensure that no persons are on, in or under the machine while it is in use.
- Keep the area around the machine free from materials and other obstacles.
- Always stop the machine if a product becomes jammed.
- Never put your hands or any other part of your body close to rotating parts while the machine is running.
- Place warning notices on the control panel of the machine while performing maintenance operations to the machine.



Never keep your hands or any other part of your body close to rotating parts while the machine is in use.



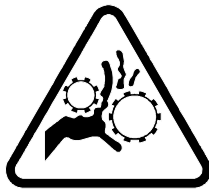
Keep the covers on the machine closed during production.

2.4 Know the pictograms

Machines manufactured by Meyn Food Processing Technology B.V. are provided with safety and protective devices. Even so, it is important to be careful when you are performing any operation on the machine. The following pictograms indicate possible dangers. The pictograms are applied to the machine and are also printed in this handbook at those places where risky activities are described.



Check regularly to ensure that all the pictograms are still in the correct positions on the machine. If any pictogram has become detached or unreadable, apply a new one immediately.



Danger
Moving parts
(rotating)



Danger
Moving parts
(up and down)



Danger
Warning



Danger
Sharp parts

3 Installation

This chapter gives a description of the actions that must be performed before the machine can be used.

3.1 Installing the Meyn vent cutter

When the machine is delivered, the adjustment of the movable parts of the machine has already been done at the factory. In chapter 7, Disassembly/assembly and adjustments, is given the information about disassembly and assembly of parts you must do to ensure the machine performs optimal.

The machine has to be placed level and stable. If necessary the machine can be adjusted by means of the adjustable legs.

3.1.1 Required floor area

Floor area required: The surface area (L*W) occupied by the machine is given in paragraph 1.1 "Technical specifications". Nevertheless, it is desirable to install the machine on a floor area that makes it easy to clean and maintain. Meyn advises you to keep a clear space of 1 meter on all sides of the machine.

3.2 The connection of the Meyn vent cutter

After installing the vent cutter, the water must be connected.

4 Operation

This chapter gives a description of the actions that must be taken before operation.

4.1 The manual controls

- The start/stop switches of the Vent cutter is located on the central control panel.
- The main switch of the overhead conveyor system is located on the central control panel.

4.2 Preparing for the starting-up procedure

- Before starting the Vent cutter clean the machine thoroughly with a high pressure waterjet to remove any dust or dirt.
- Ensure that all adjustments are correct (see Chapter 7, Disassembly/assembly and adjustments).
- Check for damage. Repair if necessary.
- Check that all emergency stop push buttons function correctly.
- Open the water supply using the ball valve.
- Start the high pressure pump that supplies the high pressure water.
- The Vent cutter is now ready for production and can be started with the rest of the evisceration line.



When starting the overhead conveyor make sure the carrier pins are positioned in the holes of the drive wheel.
If not, the height of the machine should be adjusted by means of the height adjustment before starting the line.

4.3 Start-up procedure

The Vent cutter is part of the conveyor system. This is controlled from the main control panel. On the panel is the start/stop switch of the Vent cutter. The switch is a selection switch; 1 = Start, 0 = Stop After selecting "1" the Vent cutter starts running and is ready for production.

4.4 In production

When the machine is correctly adjusted it will continue to function independently. While producing with the Vent cutter, regularly check the result of the machine see also chapter 7.

4.5 The stop procedure

The start/stop switches of the Vent cutter is located on the central control panel. Before stopping the machine it is advised to stop the supply of birds. Rotate the start/stop switch to "0" to stop the machine. The total machine will stop. Never use the emergency stop for a regular stop.

4.6 The emergency stop

For use in emergency situations, emergency stop push buttons are mounted on the overhead conveyor and on the frame of the Vent cutter. When these red push buttons are pressed, the machine will stop immediately. They may only be used in cases of emergency. If it is desired to re-start the machine after an emergency stop, the emergency stop push button must be pulled out and rotated clockwise at the same time. Thereafter, the machine can be re-started by pressing the start push button. The emergency button may also have a small button on its side. This reset button must first be depressed, before the start button can be used.

There is an emergency stop pull cord around the machine. This may only be operated in an emergency. If an operator is accidentally caught up in the machine or falls into the machine or intentionally operates it manually, the emergency stop will automatically activate and stop the entire machine. You can also activate the emergency stop by pushing the red push-button. The emergency stop pull cord and possibly the red push-button will then have to be reset.

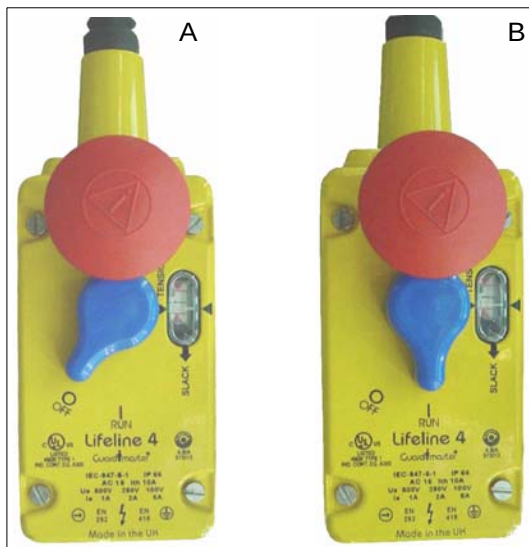


Figure 4.4 The emergency stop pull cord.

Functioning

- The blue switch is switched automatically to "0; OFF" (A) when the pull cord has been activated. See figure.
- Check the tension of the pull cord. See "Tension of the pull cord" and figure below.
- Check that the emergency situation has been resolved.
- Switch the blue switch to "RUN" (B) to reset the emergency stop pull cord. See figure.

- Reset the emergency stop switch circuit by means of pushing the lighted blue reset emergency stop button on the main panel. After this the light in the button goes off.
- Accept the fault on the touch screen.
- After this you can restart the machine.

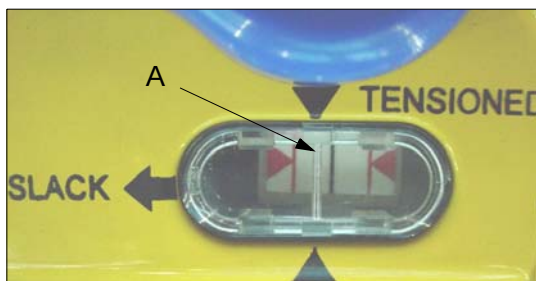


Figure 4.5 The pull cord tensioner window.

Tension of the pull cord

- The stripe (A) must be in the middle of the window.
- If necessary change the tension of the cord by turning the cord tensioner.

5 Cleaning

This chapter gives a description of the cleaning activities that must be carried out to keep the machine in good condition and to facilitate safe and hygienic operation.

5.1 Cleaning preparations

- During cleaning you are advised to wear detergent- and disinfectant proof clothing. Protect your eyes and skin.
- Ventilate the room in which you are working.
- After production, all products should be removed from the room in which the machine is installed. Remove coarse dirt.



Switch off the main switch of the machine before disassembly for cleaning.

5.1.1 Disassembly/assembly for cleaning

The Meyn Vent cutter is so designed that daily cleaning can be performed without disassembly.

5.2 General cleaning schedule

<i>During breaks</i>	<i>Daily after production</i>	<i>Monthly *</i>
High-pressure water cleaning.	High-pressure water cleaning. Alkaline foam cleaning. High-pressure water cleaning. Disinfection. High-pressure water cleaning.	Acid foam cleaning.

Table 5.1 The cleaning schedule of the machine.

*) or as often as necessary.

- Clean the machine from top to bottom.
- During cleaning, pay extra attention to:
 - Inside the machine.
 - The drill rods and vent-knives.
 - Water sprayers.
- During cleaning, never point the high-pressure water jet to the control panel or motor.
- For proper functioning, after cleaning the moving parts of the machine should be sprayed with a vegetable oil.

5.2.1 High-pressure water cleaning

With high-pressure water cleaning nearly all visible dirt will be sprayed off the machine. This cleaning is a preparation for the daily foam cleaning and disinfection. Furthermore, a high-pressure water jet can be used for localized cleaning during production breaks.

- High-pressure water cleaning should always be performed as close to the machine as possible. Stop the machine before starting the high-pressure water cleaning
- Prevent people from getting in the way of the high-pressure jet. Always hold the spray gun with both hands.

For high-pressure cleaning use the following equipment:

- A fixed piping system, delivering water-pressure with a minimum of 3,000 kPa and a maximum of 6,000 kPa and with a water temperature of approx. 55°C. at the spray-mouth
- A spray lance with a downward bent nozzle. Such a spray lance has the advantage of not transporting the dirt upwards
- A spray nozzle of the flat-jet type with a spray-angle of 15° to 25° and a water consumption of 800 to 1000 litres/hour at a water pressure of 6,000 kPa

A water temperature of 55°C is advised, since a higher temperature may cause sticking of proteins and with a lower temperature the fat will not be removed properly. The pressure should not be too high, to avoid unnecessary mist, and to protect e.g. bearings.

The unit kPa (kilopascal) is SI standard for pressure:

<i>New</i>		<i>Old</i>
100 kPa	14,5 PSI	1 bar
3000 kPa	435 PSI	30 bar
10000 kPa	1450 PSI	100 bar

Table 5.2 **SI.**

5.2.2 Alkaline foam cleaning

Alkaline foam cleaning ensures that fat, proteins etc. will be degraded chemically.

- Foam the machine from top to bottom. Foaming may be performed while the machine is running slowly, provided sufficient distance is kept to the moving parts of the machine. Mind the other machines.
- After foam cleaning remove the dirt and foam residue thoroughly by means of the high-pressure water jet.

For alkaline foam cleaning use the following equipment:

- A fixed piping system for foam delivery.
- A large spray-nozzle of the flat-jet type, through which the foam can be distributed equally along the surface.
- The alkaline foam detergent used should not affect the materials which it will contact. Consult your detergent supplier to know which detergents will fulfil these specific requirements and which cleaning directions to follow. Obey these instructions. Incorrect application may be dangerous

Please avoid:

- A too high concentration.
- A too long contact
- Not rinsing
- Mixing with other products

It is advisable not to use a combination of detergents and disinfectants ('detergent sanitizers'). This leaves the machine too dirty to be disinfected effectively.

A combination of these two methods is of no use at all.

5.2.3 Disinfection

Alkaline foam cleaning alone does not lead to a complete killing and removal of bacteria, fungi, viruses, algae and yeasts. For that purpose a separate disinfectant is needed. Disinfect the machine daily, after the alkaline foam cleaning.

- Disinfection only makes sense when the surfaces are visibly clean after the alkaline foam cleaning.
- Spray the disinfectant onto the machine and surrounding area. When disinfecting, it is difficult to judge just how thorough you have been.
- We advise you to disinfect the machine and the area immediately surrounding it in an established pattern.
- After applying, the remaining foam rests should be removed by means of high-pressure water jet.



Damage may be caused by not thoroughly rinsing the disinfectant. Evaporation of chlorous drops will lead to a very high concentration of chlorine. In combination with long contact-times can cause corrosion to metals.

For disinfection use the following equipment:

- A fixed piping system.
 - A spray lance with downward-bent top.
 - A spray nozzle of the flat-jet type, with a spray angle of 15° to 25°.
 - The used disinfectant should not affect the materials which it will contact.
- Consult your detergent supplier to know which detergents will fulfill these specific requirements and which cleaning directions to follow. Obey these instructions. Incorrect application may be dangerous.

5.2.4 Acid foam cleaning

"Hard" water causes calcium and rust deposits on the machine. In order to remove these, a regularly performed acid foam cleaning is necessary. Depending on the hardness of the water this may be either once a month or more often if necessary.

- Foam the machine from top to bottom. Foaming may be performed while the machine is running, provided sufficient distance is kept from the moving parts of the machine. Mind the other machines and the overhead conveyor.
- After affecting, the dirt and foam residue should be removed thoroughly by means of the high-pressure water jet.



Contact between acids and chlorous products results in a poisonous gas. Ensure that these products do not come in contact with each other. Ventilate the room in which you are working.

For acid foam cleaning use the following equipment:

- A fixed piping system for foam supplying.
- A large spray-nozzle of the flat-jet type, that distributes the foam equally along the surface.
- The used acid foam detergent should not affect the materials with which it comes into contact. Consult your detergent supplier which detergents fulfil these specific conditions and the concerning directions for operation. Obey these instructions carefully. An incorrect application may be dangerous.

It is advisable not to use spring water for the production of foam; this water may contain a high degree of chloride. Tap water reduces the chance of rust.

5.3 Advising cleaning chemicals to the customer.

Customers sometimes ask Meyn Food Processing Technology B.V. to advise them about cleaning chemicals and disinfection chemicals. As a result of these questions some recommendations are as follows as stated below.

We recommend:

- To contact the major manufacturers or suppliers of these chemicals, like Henkel, Diversey etc. These companies have a good knowledge of cleaning chemicals and their corrosive or destructive properties on materials.
- Do not buy unknown (cheap) products.
- To follow all user instructions carefully.
- Meyn Food Processing Technology does not recommend to use specific cleaning chemicals, because we do not know which chemicals are available in each specific country. We can not know the (secret, complex) composition of the products. Besides this, Meyn Food Processing Technology does not accept any responsibility for any damage caused by wrong application of cleaning chemicals.
- The above mentioned can only be avoided by a direct contact between user and supplier of cleaning chemicals.

6 Maintenance and trouble shooting table

This chapter describes the maintenance activities that are necessary to keep the machine in good condition and to facilitate safe operation. We advise you to maintain the machine regularly. This lengthens the working life of the machine and reduces the chance of failures.

6.1 Lubricants

Lubricants used in the food processing plants are categorized and registered by the NSF in two groups:

- H1 lubricants, are used when incidental contact with food products cannot be excluded.
- H2 lubricants, are used for all other applications in a food plant.

For all bearings and moving parts Meyn advises to use a multi purpose NLGI 2 H1 grease with the following characteristics;

- Excellent water resistance.
- Chemical resistant.
- Food grade H1.
- High pressure (EP) properties.
- Compatible with other greases.

It is advised not to mix greases!

Meyn is using “Jax Halo Guard FG2” as a general purpose grease for most applications. Other greases are used for:

- For high speed rotations (> 5000 rpm).
- For low temperatures (< - 5°C or 23 F).
- For gearboxes.
- For Hydraulics.

For the hydraulic height adjustments of ECP evisceration machines Hydraulic Fluid 32 Food grade H1 is used.

Meyn is using “Rocol Foodlube multi paste” in order to mount threaded stainless steel parts like bolts and nuts together, this special paste is used that prevents seizing of the threaded parts and allows for disassembly when needed.

6.2 Checklist

<i>Part</i>	<i>Check</i>
Daily, All adjustments	Check height, inlet guide and output from the machine.
Daily, performance	Check the output of the machine.
Daily, Sprayer	Check the holes in the sprayer head for stoppages. If the holes in the sprayer head are blocked by stoppages, prick them open with a sharp object.
Weekly, Torque spring	Check for broken spring. Replace the broken springs.
Weekly, Compression springs	Check for broken springs. Replace the broken springs.
Weekly, vent knives.	Check for sharpness and damage. Replace if necessary.
Monthly, sleeve-bearing bush	Check for wear. If necessary replace the sleeve-bearing bush.
Yearly, Bearings on slide blocks.	Replace.
Yearly, Main bearing	Check clearance. If necessary replace the main bearing.

Table 6.1 Checklist

6.3 Lubrication-point list

<i>Part</i>	<i>lubricant</i>
Daily after cleaning, Slide shafts.	Vegetable oil.
Daily after cleaning, The drill units	Vegetable oil.
Every two weeks, Bearings drive wheel.	See section 6.1
Every two weeks, Bearings upper plate.	See section 6.1
Every two weeks, Bearings lower plate.	See section 6.1.
Every two weeks, Bearings lift-up drum.	See section 6.1.

Table 6.2 Lubrication-point list

6.4 Trouble shooting table

<i>Failure</i>	<i>Cause</i>	<i>Solution</i>
Machine doesn't start.	• Emergency stop push button has been pressed. • No power supply. • Machine too low. The carrier pins of the main shaft are not positioned in the holes of the driving wheel.	• Resolve the emergency situation and reset the emergency stop. • Insert plug socket. • Raise the machine.
Machine doesn't cut.	• Vent knives are blunt or broken. • Height of the bottom cam is sagged. • Height adjustment of the machine is incorrect.	• Sharp the blunt knives and replace the broken vent-knives. • Adjust the height of the lift-up drum correctly. • Adjust the height of the machine correctly.
Missing a bird.	• Input not correct. bird is positioned between the units instead of in the unit. • Positioning in the unit is not correct, vent is not positioned under the drill. • Machine too low.	• Timing shackle with regard to the unit not correct. • Put the back plate in the proper position (vent of the bird exactly below the vent knife) • Raise the machine.
Breaking the intestine.	• Vent-knives are blunt.	• Sharpen or replace the vent-knives.
Damage to the intestines.	• The vent-knife cuts too deep. • Probes not adjusted right.	• Lower the position of the bird. • Adjust probes.
Back or kidneys damaged.	• Birds are not correctly positioned.	• Put the machine into the correct position.

<i>Failure</i>	<i>Cause</i>	<i>Solution</i>
Hips cut.	• Diameter of the vent knife is too big.	• Change the vent knife.
Vents attached to the birds.	• Probes not clean • Vent knives not sharp. • Not cutting deep enough. • Back plates out too far. • Probes not adjusted right.	• Adjust high-pressure nozzle. • Check high pressure pump (filter). • Replace or sharpen the vent knife. • Raise bottom cam of the machine. • Adjust back plate cam. • Adjust probes deeper.
Bursa not removed.	• Probes not clean. • The vent-knives not sharp. • Back plates not far enough out. • Back plate cam not in right position.	• Adjust high pressure nozzle. • Check high-pressure pump. • Replace or sharpen the vent-knives. • Adjust back plate cam. • Adjust back plate cam.
A lot of vents pulled off birds.	• Not cutting deep enough. • Back plates out too far. • Back plate cam not in right position.	• Raise bottom cam . • Adjust back plate cam. • Adjust back plate cam.

Table 6.3 Trouble shooting table.

7 Disassembly/assembly and adjustments

This chapter describes how to disassemble, assemble and adjust components of the machine after maintenance and/or replacement of parts.

7.1 Basic adjustments

The default settings to the various components of the units in relation to each other are carried out in the factory. However, as a result of adjustments to the parts and for local circumstances, it may be necessary to re-adjust the default settings.



Switch off the main switch of the machine before cleaning/adjustments.

Tip!

Put marks on the parts to be readjusted before changing the basic settings. This will save you a lot of time and annoyance.



Place warning signs with the control panels of the machine and the overhead conveyor.

7.2 Adjustments with machine not running

7.2.1 The guide bars.

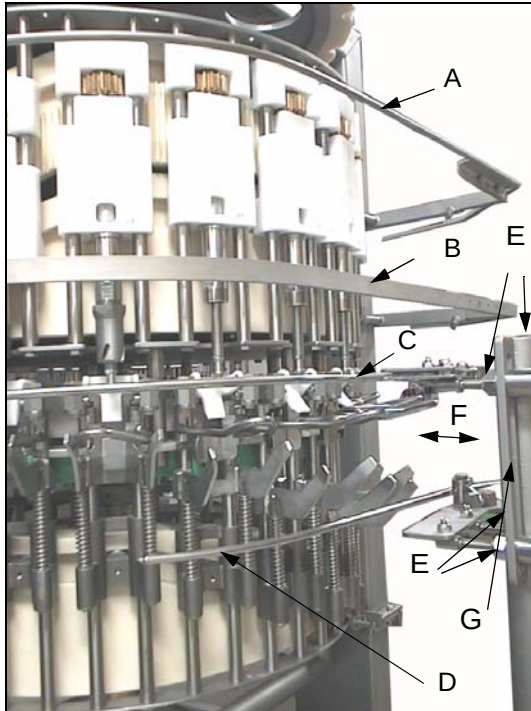


Figure 7.6 Position of the guide bars.

- Guide bar (A) keeps the shackles from touching the brass gears and keep the bird in right position.
- Guide bar (B) keeps the shackles from touching the blade units.
- Guide bar (C) guides the birds properly in the lifting unit. Adjust guide bar (C) in height there where the leg loops are just not touching it when the bottom part of the machine is in the highest production setting.
- Guide bar (D) guides the birds properly in the lifting unit. Adjust guide bar (D) in height where the lifting units are just not touching it when the bottom part of the machine is in the lowest position.

The in feed bars (C and D) should be adjusted close enough to the machine to guide the birds properly in the lifting unit. Check these guide bars regularly. The adjustment of the in feed guide bars and height of the machine is good if most of the tails are grabbed by the tail clips. This can be checked at the in feed side just before the blades are starting to cut the vents.

Vertical adjustment:

- Loosen the nuts (E).
- Adjust the bars in height by means of the sleeves (G).
- Tighten the nuts (E).

Horizontal adjustment:

- Loosen the serrated knobs.
- Put the guides in the desired position.
- Tighten the serrated knobs.

7.2.2 Adjust the Cams

Five cams (top cam, bottom cam, tail clip cam, back plate cam and knife unit cam) in which a curve path is milled or grooves are milled drive the processing units.

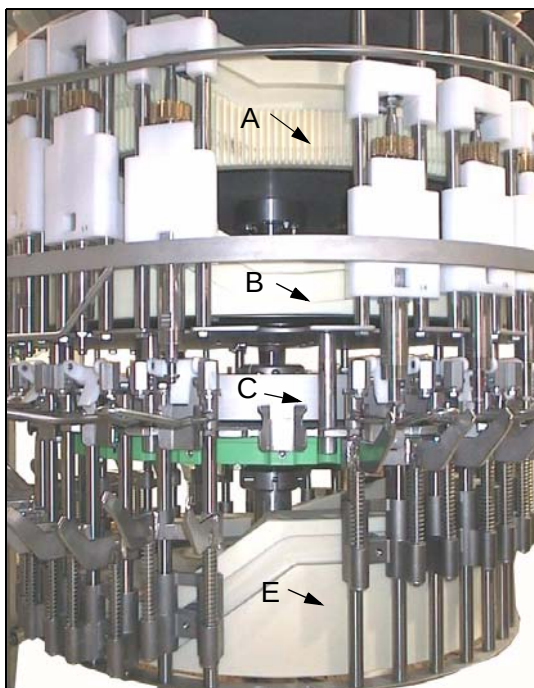


Figure 7.7 View cams.

- Top cam (A)
- Knife unit cam (B).
- Tail clip cam (C).
- Back plate cam (D).
- Bottom cam (E).

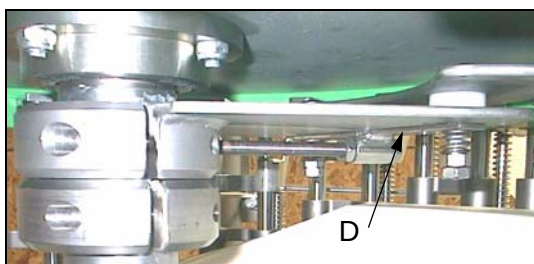


Figure 7.8 View back plate cam.

- Back plate cam (D).

7.2.2.1 Top cam

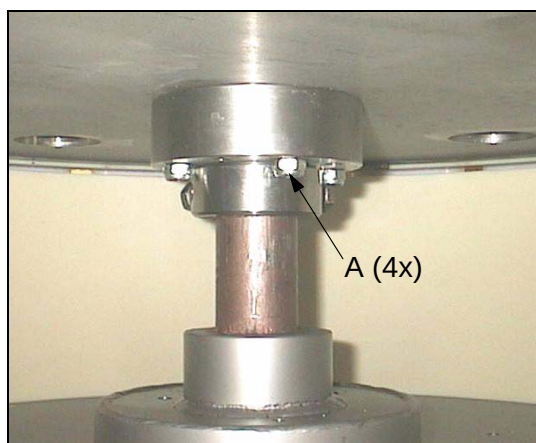


Figure 7.9 Adjust the top cam

Adjust the top cam.

- Remove one top unit to allow access to the bottom of the cam. See section 7.2.10.
- Loosen four nuts (A).
- Turn the cam in the desired position.
- Tighten the nuts.

7.2.2.2 Knife unit cam

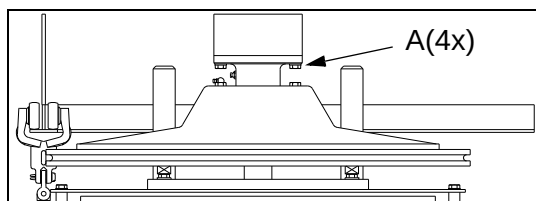


Figure 7.10 Adjust the knife unit cam.

Adjust the knife unit cam.

This cam is mounted on the main shaft with a key. In order to change the position of this cam the main shaft has to be turned.

- Loosen four bolts (A).
- Loosen four nuts (B).
- Turn the cam in the desired position.
- Tighten the bolts (A) and nuts (B).

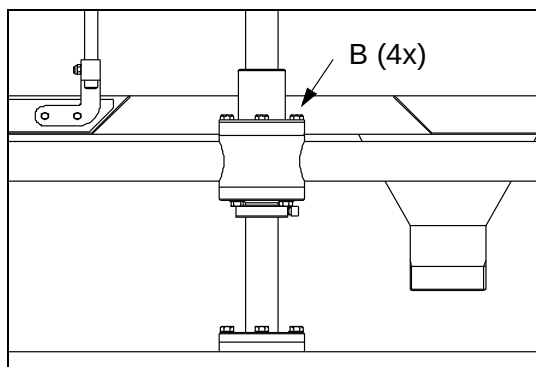


Figure 7.11 Adjust the knife unit cam.

7.2.2.3 Tail clip cam.

Remove one tail clip unit and lower bottom part of machine to allow access to the cam. See section 7.2.10.

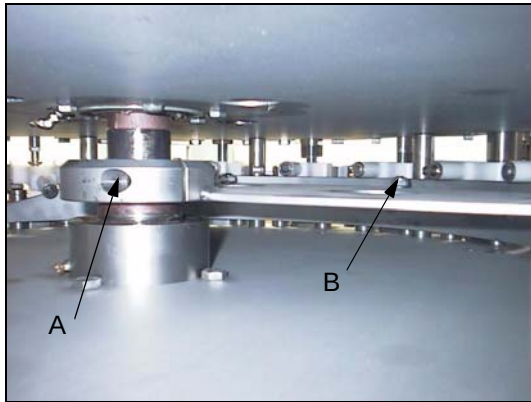


Figure 7.12 Adjust the tail clip cam.

Adjust the tail clip cam

- Loosen three nuts (B).
- Turn the cam in the desired position.
- Tighten the nuts

Or if the cam won't turn (end of sleeves)

- Loosen two allen bolts (A).
- Turn the cam in the desired position.
- Tighten the bolts.

7.2.2.4 Back plate cam

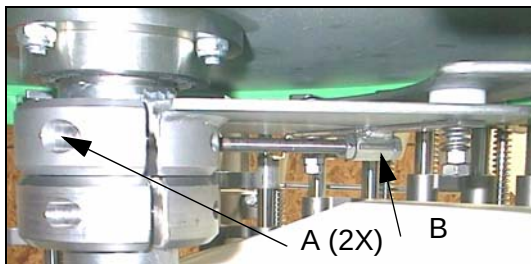


Figure 7.13 View back plate cam.

Adjust the timing

Back plates slide out sooner or later.

- loosen two allen bolts (A).
- Turn the cam in the desired position.
- Tighten the bolts (A).

Change the distance

To change the distance of the back plates

- Move bolt (B) clockwise: The back plate will slide out less.
- Or move bolt (B) counter clockwise: Back plate will slide out more.

7.2.2.5 The bottom cam

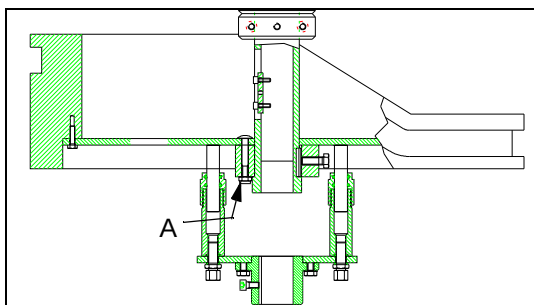


Figure 7.14 Adjust the bottom cam.

Adjust the bottom cam.

- Loosen four nuts (A).
- Turn the cam in the desired position.
- Tighten the nuts (A).

7.2.3 Position of the cams

7.2.3.1 In feed side

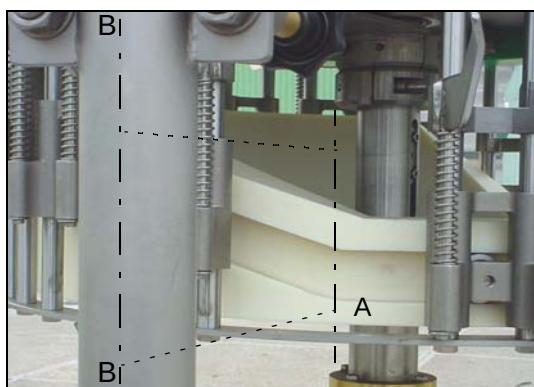


Figure 7.15 Out feed side

- The bend (A) in the curve must be in line with the centre line (B) of the water pipe.

7.2.3.2 Out feed side



Figure 7.16 In feed side

- The bend in the curve of the three cams must be in line (A) with each other.

7.2.3.3 Back plate cam

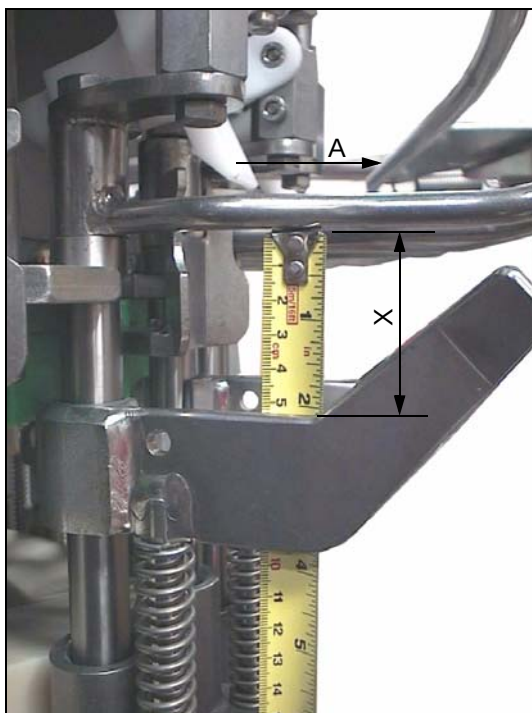


Figure 7.17 Back plate cam

Back plate cam

- The back plate should start to come out (A) when the distance leg loop, top lifting unit (X) is approximately 2 inch (51 mm).

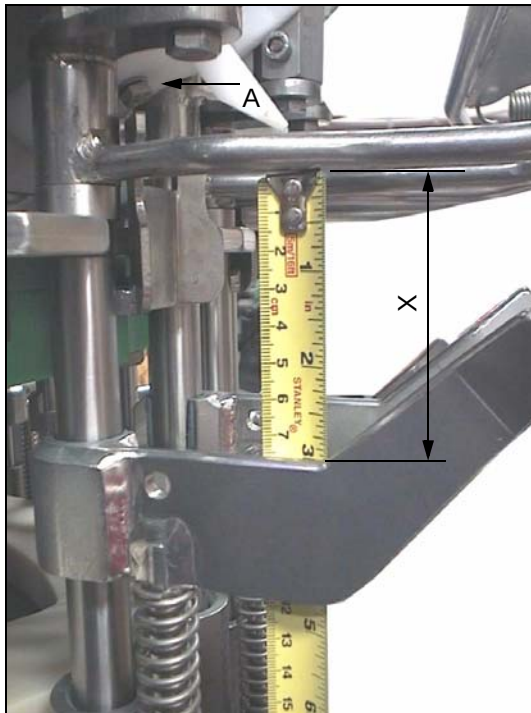


Figure 7.18 Tail clip cam

Tail clip cam

- The tail clip should start to go in when the distance bottom leg loop: Top lifting unit is approximately 3 inch (76 mm).

7.2.4 The in feed

The shackles of the overhead conveyor need to line up with the units of the machine to get a good in feed of the birds.

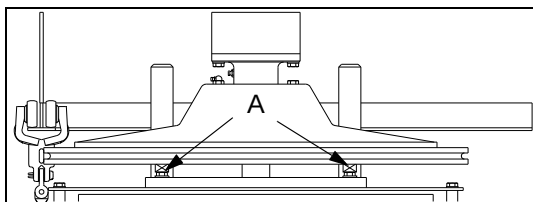


Figure 7.19 Adjust the blade unit cam.

- loosen six nuts (A).
- Turn the machine in the desired position.
- Tighten the nuts (A).

7.2.5 The probes (vent cone)

All the probes have to be set like figure below. Distance X needs to be approximately 1/16 inch (1,5 mm.). When the probes are set too low there will be too much attached vents. Probes set too deep can cause damaged intestines.

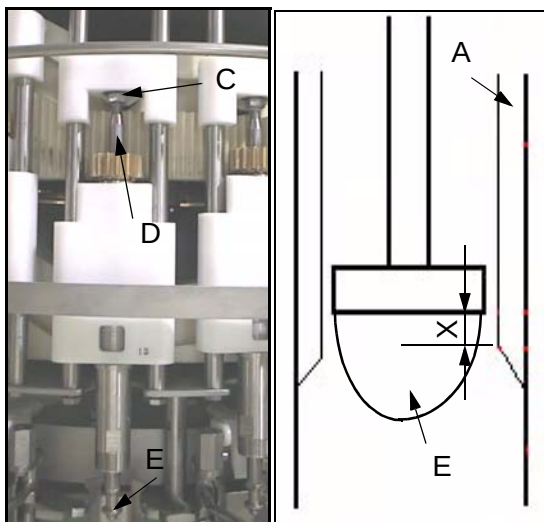


Figure 7.20 Adjustment of the probe

Adjust the probes at the out feed side of the machine (there were the knife unit is going up).

- Loosen nut (C).
- Turn shaft (D) right to move probe (E) deeper in nylon clamp bush (A).
- Turn shaft (D) left to move probe (E) less deep in nylon clamp bush (A).
- Tighten nut (C).

7.2.6 Vent knife

A vent-knife cuts the vent and bursa of the bird. For optimal cutting performance the vent-knives must be sharp. These knives must be sharpened regularly (after two to four weeks). They have to be removed from the drill unit for that purpose. New vent knives are 60 mm long. They can be sharpened until a length of 55 mm remains. When a vent-knife is broken or is too short, you have to replace it.

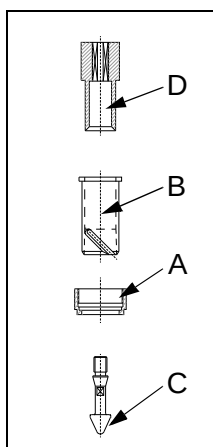


Figure 7.21 Drill unit (principle).

Replace a vent-knife

- Loosen the nut (A).
- Remove the vent knife (B).
- Loosen the vent cone (C).
- Replace the clamp bush (D).
- Replace the vent cone (C).
- Replace the vent knife (B).
- Tighten the bolt (A).

Five sizes of vent knives can be mounted on the drill unit. When you change the vent knife size, the vent cone and clamp bush have to be replaced also. The table below shows the order numbers for ordering of the knives, cones and bushes. Please mind the different numbers for left and right turning machines.

Vent knife size (MM)	Machine type	Vent knife (B) n.	Vent cone (C) n.	Clamp bush (A) n.
Ø 22.2	Right turning	0338.0000.289.00	0595.0000.036.00	0595.0000.017.00
Ø 22.2	Left turning	0338.0001.284.00	0595.0000.036.00	0595.0000.017.00
Ø 25.2	Right turning	0338.0000.006.00	0595.0000.036.00	0595.0000.030.00
Ø 25.2	Left turning	0338.0001.006.00	0595.0000.036.00	0595.0000.030.00
Ø 29	Right turning	0338.0000.281.00	0595.0000.038.00	0595.0000.022.00
Ø 29	Left turning	0338.0001.332.00	0595.0000.038.00	0595.0000.022.00
Ø 32	Right turning	0338.0000.316.00	0595.0000.038.00	0595.0000.022.00
Ø 32	Left turning	0338.0001.331.00	0595.0000.038.00	0595.0000.022.00
Ø 33,5	Left turning	0338.0000.317.00	0595.0000.038.00	0595.0000.022.00
Ø 33.5	Right turning	0338.0001.333.00	0595.0000.038.00	0595.0000.022.00

7.2.7 Position of the back plates

The distance should approximately 3/4 inch (19,05 mm) for 6 pound (2,721 kg) bird and approximately 1 1/4 inch (31 mm) for a 4 pound (1,80 kg) bird.

The position of the back plates does normally not need to be changed for different growers.



Figure 7.22 Position of the back plates.

- If the back plates are out too far the knife will drill in the back of the birds what can cause attached vents and many vents can be pulled of the intestine.
- If the back plates are not cut out far enough the bursa of fabricius won't be cut out (bursa).

7.2.8 High pressure nozzle

The first high-pressure nozzle (seen from the in feed side) needs to spray the vents of the probes (vent cones). The second nozzle is to clean the knife.

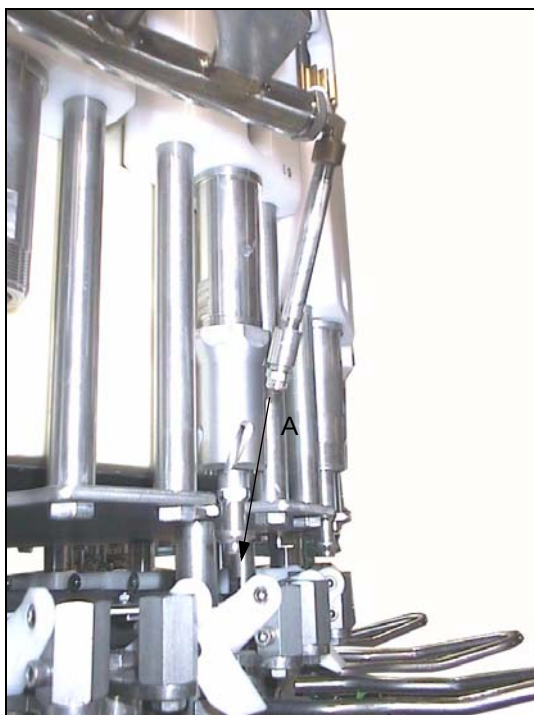


Figure 7.23 High-pressure nozzle

Aim the nozzle there where it sprays the vent of the probe just after the probe comes out the vent-knife (A).

If the vents are not sprayed of the probes by the first nozzle there will be more vent pulled off.

If not all the vents are sprayed of the probes after the second nozzle the performance of the machine will be bad.

7.2.9 Length of the intestines

The length of the intestines X can be changed by changing the position of the main shaft (see section about: adjust the knife unit cam).

- If X is too long more intestines get tangled around other machines in the evisceration line.
- If X is too short there is a change that the opener will cut the vent off.
- Length X should be approximately 11/2 inch (31,75 mm). Normally this setting is made with start-up of the machine and is not changed anymore.

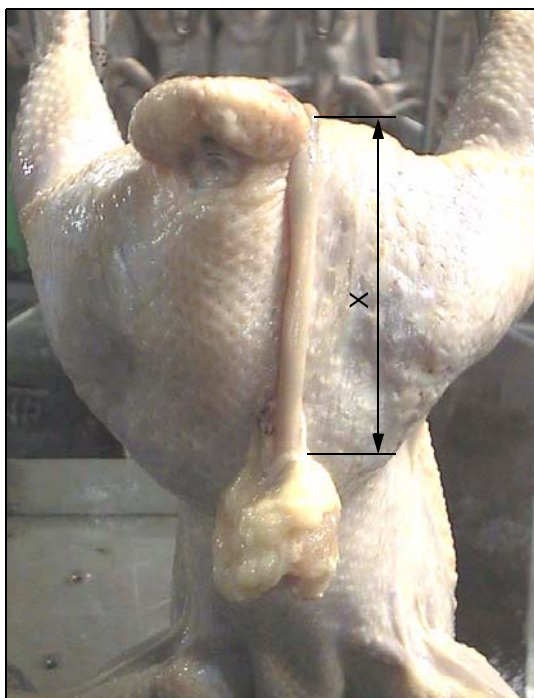


Figure 7.24 Length of the intestines.

- To make X shorter the shaft needs to be turned against the running direction of the machine.
- To make X longer the shaft needs to be turned with the running direction of the machine.

7.2.10 Removing units

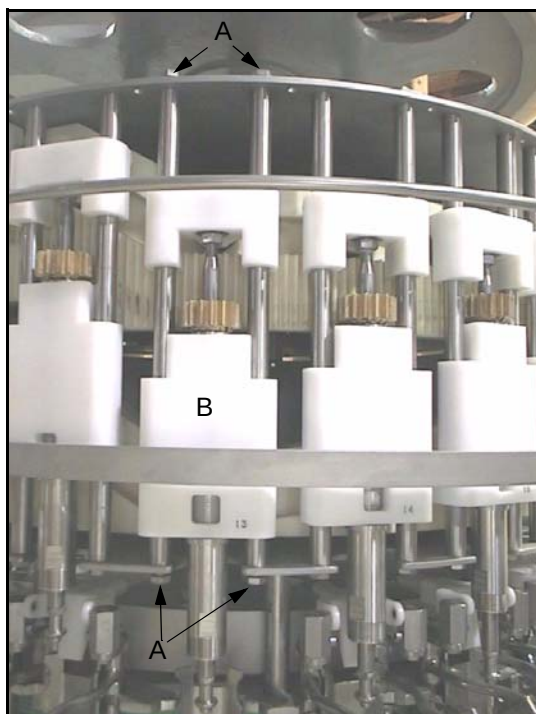


Figure 7.25 Remove a top unit.

Top unit

- Remove four bolts (A).
- Remove.
- Replace four bolts (A).



Figure 7.26 Remove a tail clip

Tail clip unit

- Remove two bolts (C).
- Remove tail clip.



Figure 7.27 Remove a bottom unit

Bottom unit

- Remove four bolts (D).
- Remove unit.

7.2.11 Torque switch (if applied)

The torque switch is a machine protection.
If the machine hangs-up, this switch will stop the evisceration line.

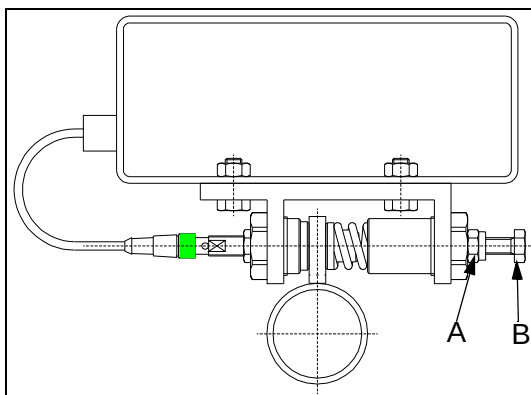


Figure 7.28 Torque switch.

Adjust the Spring tension

If the machine is not turning stable it might be necessary to change the spring tension.

- Loosen nut (A).
- Turn bolt (B) to obtain the desired spring tension.
- Tighten nut (A).

7.3 Adjustments with machine running

7.3.1 Height of the Machine.

The height of the machine is adjustable by using a handwheel (A). The height of the curve is adjustable by using a capstan wheel (B).

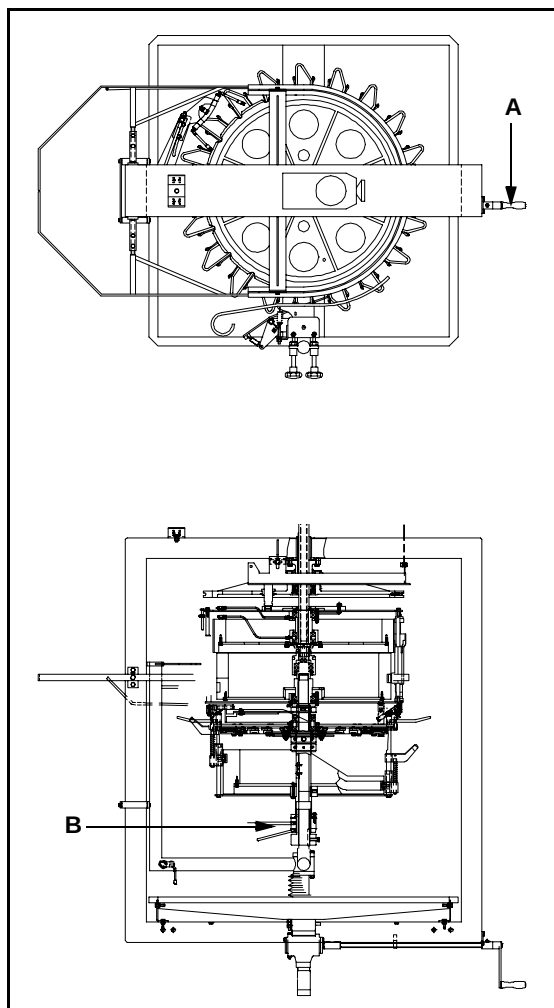
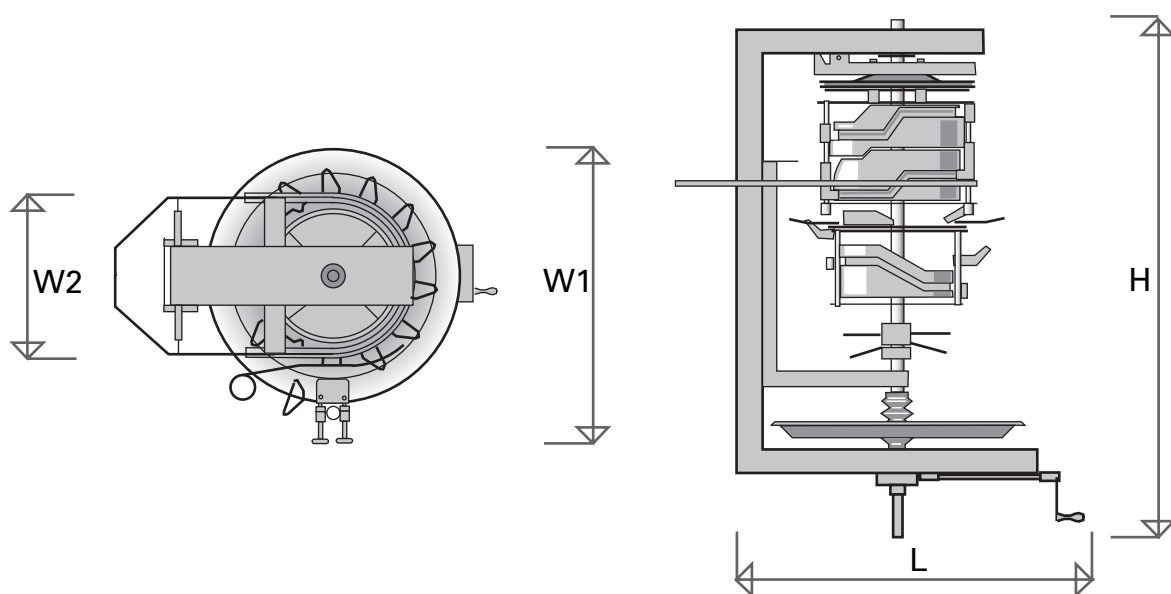


Figure 7.29 Height adjustments.

Meyn vent cutter



Capacity	BPH	3,000	4,500	6,000	7,500	8,000	10,000
Live weight	kg	0.5-3.8	2.5-5.0	0.5-3.8	2.5-5.0	0.5-3.8	0.5-3.8
Number of units		7	11	14	15	16	20
Shackle pitch	inch	12"	8"	6"	8"	6"	6"
Lenght (L)	mm	1,745	1,745	1,745	2,335	1,760	2,335
Width (W1)	mm	1,295	1,295	1,295	1,640	1,345	1,640
Width (W2)	mm	680	710	680	970	780	970
Height (H)	mm	2,150	2,150	2,150	2,150	2,150	2,150
Weight	kg	405	600	600	720	650	850
Water connection	BSP	3/4"	3/4"	3/4"	3/4"	3/4"	3/4"
Water consumption	m ³ /h	0.6	0.6	0.6	0.6	0.6	0.6
Drain connection	BSP	DN100	DN100	DN100	DN100	DN100	DN100
Options:							
High pressure pump	kW	2.2	2.2	2.2	2.2	2.2	2.2
Driven vent knife for capacity below 3,000 BPH	kW	0.55	0.55	0.55	0.55	0.55	0.55

Technical Data Sheet

Head office:

Meyn Food Processing Technology B.V.
Noordeinde 68, 1511 AE Oostzaan
P.O. Box 16, 1510 AA Oostzaan
The Netherlands

T: +31 (0) 75 - 684 3355

F: +31 (0) 75 - 684 4150

I: www.meyn.nl

MEYN

Poultry Processing Solutions

Meyn vent cutter

- Curve driven units.
- High pressure water needed only.
- Easy to adjust.

